# MARC AURÈLE GILLES

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## RESEARCH SUMMARY

I develop mathematical theory, algorithms, and software for scientific computing, with a focus on numerical linear algebra and statistical inference from noisy cryo-EM data.

## EDUCATION

**Cornell University** September 2014 - May 2019
Ph.D. in Applied Mathematics *Ithaca, NY*
Advised by Prof. Alex Townsend
Thesis title: At the intersection of differential equations and optimization: inverse problems, path planning and Krylov subspaces.

**Rutgers University** January 2012 - May 2014
B.A. in Mathematics *New Brunswick, NJ*
Summa Cum Laude
Minor in Economics

**Raritan Valley Community College** September 2010 - January 2012
Concentration in Economics *Branchburg, NJ*

## PROFESSIONAL EXPERIENCE

**Princeton University, Department of Mathematics** September 2024 – Present
*Assistant Professor of Mathematics* *Princeton, NJ*

**Princeton University, PACM** September 2021 – September 2024
*Postdoctoral Research Associate* *Princeton, NJ*
Mentor: Prof. Amit Singer.

**Facebook Reality Labs** July 2019 – September 2021
*Research Scientist* *Redmond, WA*

- Developed efficient algorithms for precise calibration, processing, and state estimation for novel sensors and imaging technologies.
- Invented and patented novel sensors for augmented reality.
- Mentored and managed two Ph.D. interns on research projects.

**Facebook Reality Labs** June 2018 – August 2018
*Research Intern* *Redmond, WA*

- Designed computer vision and optimization algorithms for eye tracking and calibration.

**Argonne National Laboratory** June 2017 – August 2017
*Research Intern* *Lemont, IL*

- Developed massively parallel C/MPI code for 3D X-ray reconstruction, solving inverse problems with tens of millions of unknowns on tens of thousands of cores.

## TEACHING EXPERIENCE

**Princeton University** January 2023 - Present
*Instructor* *Princeton, NJ*

- MAT 321/APC 321: Numerical Analysis and Scientific Computing (Fall 2024, Fall 2025, Fall 2026)
- MAT 321/APC 321: Numerical Methods (Spring 2023)
- MAT 204: Advanced Linear Algebra with Applications (Spring 2025, Spring 2026)
- Mathematics Junior Seminar: "Approximation Theory and Approximation Practice" (Spring 2026)
- Mathematics Junior Seminar: The Top 10 Algorithms of the 20th Century (Fall 2024)

**Cornell University** September 2014 - July 2019
*Teaching Assistant* *Ithaca, NY*

- Classes taught:
  INFO 2950 - Introduction to Data Science (Head TA)
  CS4780/5780 - Machine Learning for Intelligent Systems
  CS 1112 - Introduction to Computing Using MATLAB
  MATH 1910 - Calculus for engineers
  MATH 1920 - Multivariable Calculus for engineers
  MATH 1110 - Calculus I

## ADVISING

### Postdoctoral researchers

- Yuguan Wang (co-advised with Amit Singer), Postdoctoral Research Associate, Princeton University (2026 - present). Extending RECOVAR to cryo-electron tomography

### Graduate students

- Siji Chen, Ph.D. student, Program in Applied and Computational Mathematics, Princeton University (2025 - present)

### Undergraduate students

- James Li, Princeton Undergraduate (Class of 2029). Summer research "Noise-robust variational autoencoders for cryo-EM heterogeneity analysis"
- Derek Wang, Princeton Undergraduate (B.A., Mathematics, 2028). Summer research on randomized numerical linear algebra
- Ryan Divan, Princeton Undergraduate (B.A., Mathematics, 2027). Research on randomly pivoted low-rank approximation; co-author of [1]
- Matias Andia, Princeton Undergraduate (B.A., Mathematics, 2027). Research on "Functional tensor decompositions by recursive Gaussian elimination", preprint forthcoming.
- Brandon Maher, Princeton Undergraduate (B.A., Mathematics, 2026). PACM certificate - project on "Linearly transformed sparse PPCA"
- Diego Rubio Garcia, Princeton Undergraduate (B.A., Mathematics, 2026). Senior thesis on randomized low-rank matrix completion for cryo-EM heterogeneity analysis
- Iniabasi Ekpenyong, Princeton Undergraduate (B.A., Computer Science, 2026). Summer research "Neural network-based solutions to high-dimensional PDEs"
- Won-Jae Chang. Princeton Undergraduate (B.A., economics, 2024). PACM certificate - project on "Efficient algorithms for Ewald sphere correction in cryo-EM"

## HONORS AND PRIZES

· McGraw Center's examplar mentor award for graduate students and postdocs (2024)
· SIAM Student Travel Award (3×) in 2017 and 2018, 2023
· NSF Mathematical Sciences Graduate Internship in 2017
· Lawrence Corwin Memorial Math Prize in 2014
· Stanley E. Brasefield Mathematics Scholarship in 2013

## FUNDING

· Co-PI, "RECOVAR-ET: Advancing Cryo-Electron Tomography Heterogeneity Analysis", Chan Zuckerberg Biohub, $200,000, July 2026 - June 2027

## PUBLICATIONS AND PREPRINTS

1. Ryan Divan and Marc Aurèle Gilles. Convergence rates of randomly pivoted methods for low-rank approximation. *arXiv preprint arXiv:2609.06287*, 2026

2. Marc Aurèle Gilles. Convergence rates for pivoted QR and LU. *arXiv preprint arXiv:2607.26863*, 2026

3. Marc Aurèle Gilles and Heather Wilber. Low-rank approximation by randomly pivoted LU. *arXiv preprint arXiv:2601.22344*, 2026. In review, SIMAX

4. Luke Evans, Lars Dingeldein, Roberto Covino, Marc Aurèle Gilles, Erik Thiede, and Pilar Cossio. Counting particles in cryo-electron microscopy may result in incorrect population estimates. *Communications Biology*, 9:421, 2026

5. Miro A Astore, Geoffrey Woollard, David Silva-Sánchez, Wenda Zhou, Mykhailo Kopylov, Khanh Dao Duc, Roy R Lederman, Yilai Li, Yi Zhou, Jing Yuan, Fei Ye, Quanquan Gu, Rémi Vuillemot, Slavica Jonic, Lan Dang, Steven J Ludtke, Hannah Bridges, Serena Liu, Michael McLean, Valentin Peretroukhin, Johannes Schwab, Eduardo R Cruz-Chú, Peter Schwander, Marc Aurèle Gilles, Amit Singer, David Herreros, Jose-Maria Carazo, Carlos O S Sorzano, J Ryan Feathers, Ellen D Zhong, Nikolaus Grigorie, Pilar Cossio, and Sonya M Hanson. The inaugural Flatiron Institute cryo-EM conformational heterogeneity challenge. *bioRxiv*, 2025

6. Marc Aurèle Gilles and Amit Singer. Cryo-EM heterogeneity analysis using regularized covariance estimation and kernel regression. *Proceedings of the National Academy of Sciences*, 122(9):e2419140122, 2025

7. Andy Zhang, Oscar Mickelin, Joe Kileel, Eric J Verbeke, Nicholas F Marshall, Marc Aurèle Gilles, and Amit Singer. Moment-based metrics for molecules computable from cryogenic electron microscopy images. *Biological Imaging*, 4:e3, 2024

8. Marc Aurèle Gilles and Amit Singer. A molecular prior distribution for bayesian inference based on wilson statistics. *Computer methods and programs in biomedicine*, 221:106830, 2022

9. Eric J Verbeke, Marc Aurèle Gilles, Tamir Bendory, and Amit Singer. Self fourier shell correlation: properties and application to cryo-et. *Communications Biology*, 7(1):101, 2024

10. Marc Aurèle Gilles and Alex Townsend. Continuous analogues of Krylov subspace methods for differential operators. *SIAM Journal on Numerical Analysis*, 57(2):899–924, 2019

11. Marc Aurèle Gilles, Christopher Earls, and David Bindel. A subspace pursuit method to infer refractivity in the marine atmospheric boundary layer. *IEEE Transactions on Geoscience and Remote Sensing*, 57(8):5606–5617, 2019

12. Marc Aurèle Gilles, Youssef Nashed, Ming Du, Chris Jacobsen, and Stefan Wild. 3D X-ray imaging of continuous objects beyond the depth of focus limit. *Optica*, 5(9):1078–1086, 2018

13. Marc Aurèle Gilles and Alexander Vladimirsky. Evasive path planning under surveillance uncertainty. *Dynamic Games and Applications*, 10(2):391–416, 2020

## SELECTED TALKS AND PRESENTATIONS

- *RECOVAR: heterogeneity analysis in cryo-EM and cryo-ET*, invited seminar, MRC Laboratory of Molecular Biology, Cambridge, UK, November 27, 2026 (upcoming)
- *Gaussian elimination revisited*, Mathematical Sciences Colloquium, Rensselaer Polytechnic Institute, Troy, NY, USA, October 14, 2026 (upcoming)
- *Finding Latent Structure at Very Low SNR: Heterogeneity Analysis in Cryo-EM*, Workshop on Mathematical Structure in Data: Topology, Geometry, and Applications, Pennsylvania State University, USA, September 30–October 1, 2026 (upcoming)
- *What are the effects of the noise in heterogeneity analysis?*, CAHRA Workshop, Rutherford Appleton Laboratory, UK, September 18, 2026
- *Finding Latent Structure at Very Low SNR: Heterogeneity Analysis in Cryo-EM*, Workshop on Mathematical Foundations of AI Models, Institute for Mathematical Sciences, National University of Singapore, Singapore, August 19, 2026
- *What are the effects of the noise in heterogeneity analysis?*, Three Dimensional Electron Microscopy Gordon Research Conference, Castelldefels, Barcelona, Spain, June 2026
- *Low-Rank Approximation by Randomly Pivoted LU*, Curves and Surfaces, Saint-Malo, France, June 9, 2026
- *Low-Rank Approximation by Randomly Pivoted LU*, 27th Conference of the International Linear Algebra Society (ILAS), Virginia Tech, Blacksburg, USA, May 20, 2026
- *Low-Rank Approximation by Randomly Pivoted LU*, Flatiron Institute, New York, USA, January 2026
- *Reconstructing conformational states and inferring conformational densities in cryo-EM*, 2025 Stanford SSRL/LCLS Users' Meeting, SLAC National Accelerator Laboratory, Menlo Park, California, 2025
- *Reconstructing conformational states and inferring conformational densities in cryo-EM*, Biophysical Society Annual Meetings, CryoEM Subgroup, Los Angeles, USA, 2025
- *Reconstructing Conformational States & Densities in CryoEM with RECOVAR*, OpenEye CUP XXIV, Santa Fe, USA, 2025
- *Reconstructing Distributions of Molecules from Cryo-EM Datasets*, PACM/CSML joint Colloquium, Princeton, USA, 2025
- *Reconstructing Distributions of Molecules from Cryo-EM Datasets*, Computational Microscopy Reunion Conference 2, Los Angeles, USA, 2025
- *RECOVAR: A Bayesian framework for cryo-EM heterogeneity analysis*, Gordon Research Seminar, Barcelona, Spain, 2024
- *Reconstructing flexible proteins from cryo-EM datasets*, SIAM ALA, Sorbonne Université, 2024
- *Reconstructing flexible proteins from massive microscopy datasets*, Applied Mathematics Seminar, Yale University, New Haven, 2024
- *Reconstructing flexible proteins from massive microscopy datasets*, Applied Mathematics Seminar (online), University of Texas at Austin, 2024
- *A Bayesian framework for cryo-EM heterogeneity analysis*, CCM-CCB seminar, Flatiron Institute, NYC, 2023
- *A Bayesian framework for cryo-EM heterogeneity analysis*, Cryo-EM One World seminar series (online)
- *Cryo-EM heterogeneity analysis by regularized covariance estimation*, IAS, Gottingen, 2023
- *Cryo-EM heterogeneity analysis by regularized covariance estimation*, ICIAM, Tokyo, 2023

- *Cryo-EM heterogeneity analysis by regularized covariance estimation*, Cryo-EM summer workshop, Flatiron Institute, New York, 2023
- *High Dimensional Covariance Estimation in Cryo-EM*, SIAM MDS, San Diego, 2022
- *Heterogeneity analysis in cryo-EM*, IPAM seminar, Los Angeles, 2022
- *A Molecular Prior Distribution for Bayesian Inference Based on Wilson Statistics*, Cryo-EM seminar, Flatiron Institute, NY (online), 2022
- *Computing with subspaces generated by differential operators*, IDeAS seminar, Princeton University, 2021
- *3D X-ray imaging beyond the depth of focus limit*, SIAM Conference on Imaging Science, 2018
- *Continuous analogues of Krylov methods for differential operators*, SIAM Conference on Applied Linear Algebra, 2018
- *Continuous analogues of Krylov methods for differential operators*, Scientific Computing and Numerics seminar, Cornell University, 2018
- *Adversarial path planning*, Scientific Computing and Numerics seminar, Cornell University, 2017

## ORGANIZED EVENTS

- Co-organizer of the PACM IDeAS seminar, the weekly applied and computational mathematics seminar at Princeton University (Spring 2026 - present)
- Co-organized "Advances of regularization techniques in iterative reconstruction" minisymposium at SIAM Conference on Imaging Science (2018)
- Organized Applied Mathematics student-invited speaker series at Cornell University (2017-2019)

## SOFTWARE

- RECOVAR [6]: `https://github.com/ma-gilles/recovar`
- RELAX: `https://github.com/ma-gilles/relax`
- C++ companion library to [13]: `github.com/eikonal-equation/Stationary_SEG`
- MATLAB companion libraries to [10] : `chebfun.org/examples/ode-linear/Krylov.html`
- Python companion library `https://github.com/ma-gilles/wilson_prior`

## RESEARCH NOTES

Marc Aurèle Gilles and Eric J Verbeke. Computational problems from the 2026 3DEM Gordon Research Conference. Technical note, `https://ma-gilles.github.io/notes/GRC_problems.pdf`, 2026